

TABLE OF CONTENTS

E-240 OUTDOOR AND SUBSTRUCTURE GROUNDING

	Page No.
1. SCOPE OF WORK.....	E-240-2
2. CODES AND STANDARDS	E-240-2
2.1 National.....	E-240-2
2.2 Institute of Electrical and Electronics Engineers	E-240-2
3. OPERATING AND DESIGN CONDITIONS	E-240-2
3.1 Soil Resistivity.....	E-240-2
3.2 Fault Currents to be used in the Design	E-240-3
4. DETAILED REQUIREMENTS	E-240-3
4.1 General	E-240-3
4.2 Conductors and Joints	E-240-4
4.3 Objects to be Connected to the Grounding System.....	E-240-5
5. PERFORMANCE TESTING AND GUARANTEES.....	E-240-6
6. QUALITY ASSURANCE/QUALITY CONTROL	E-240-6

E-240 OUTDOOR AND SUBSTRUCTURE GROUNDING

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of outdoor and substructure grounding in accordance with Part 4, subsection 1.2 Summary of Work.

This specification covers the grounding of the out door structures and the grounding underneath all buildings. The Contractor shall provide and install all ground bars, cable, ground rods, heavy duty compression fittings, exothermic welds and miscellaneous materials required for the areas. The grounding grid shall be connected to the existing ground grids.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5, Codes and Standards, and the following specific standards:

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation – Indonesia).

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 Institute of Electrical and Electronics Engineers

IEEE 80 : IEEE Guide for Safety in Substation Grounding.

IEEE 665 : IEEE Guide for Generating Station Grounding

3. OPERATING AND DESIGN CONDITIONS

3.1 Soil Resistivity

The Contractor shall do soil-resistivity measurements necessary for the design of the grounding system as well as cathodic protection system.

The soil resistivity measurement shall be performed after that area has been graded to its final elevation, to establish the design parameter of the grounding

system. The procedure and locations of these measurements shall be subject to the Owner's review.

All measurements shall be made using the WENNER four-pin method. Measurements shall be taken at a representative set of locations in the Power Plant {six (6) minimum} to determine the intermediate soil resistivity using a uniform probes, the probes shall not be inserted more than 75 mm into the soil.

Deep soil resistivity readings shall be taken in at least two (2) locations. Each measurement shall be taken with the line of the probes placed at right angles to the other measurement. The current probe spacing shall be approximately 350 meters and the potential probe spacing shall be approximately 250 meters. The probe shall not be inserted more than 300 mm into the soil.

The Contractor shall make any additional soil resistivity measurements necessary to design the grounding system.

The Contractor shall be responsible for the grounding mat and grounding rods confirmation design. The final resistance to ground shall be 0.05 ohm (measured at the driest season) or less.

3.2 Fault Currents to be used in the Design

<u>Description</u>	<u>500 kV system</u>
Ultimate total fault current	40 kA*)
Ultimate earth component of fault current	13 kA*)
Number of high speed reclosures	1
Tolerable touch potential	240 V*)
Tolerable step potential	390 V*)

Note: *) The figure is an estimate, the Contractor shall calculate to confirm there figures.

4. DETAILED REQUIREMENTS

4.1 General

The Contractor shall design and build the grounding system so that design resistance and touch and step voltages of the overall grounding system are

achieved for the initial earth component of fault current. The design shall also be such that the area of the grounding system shall not require reinforcement due to increased fault current to meet the listed safety criteria.

4.2 Conductors and Joints

4.2.1 Ground Cable

All buried ground cables shall be 240 mm² copper with a commercially pure lead jacket not less than 1.7 mm thick. All cables in the area between the substation and the station, the turbine building, the boiler building, and the chimney shall be 240 mm². The buried ground cables shall be buried 600 mm below grade.

Ground riser cables for equipment supporting or containing power conductors shall be 240 mm² copper such as on the ground conductor size being used in the area. The ground riser cables for other equipment shall be 120 mm² copper or as noted on the design and standard drawings. Ground riser cables shall have a 1.7 mm thick commercially pure lead jacket or shall be 600/1000 volt cross linked polyethylene (XLPE) insulated cables.

All lead jacketed cables shall be protected from moisture entering the cable. The ends of grounding and riser cables shall be sealed when work is not being performed on the cable, as at the end of the working day. All joints in lead jacketed or buried cables or copper bars shall be sealed so that no bare copper is left exposed in the ground.

4.2.2 Ground Rods

All ground rods shall be 20 mm diameter stainless steel rods at least 6 meters long. Ground rods shall be used at locations that are subject to impulse currents, such as arresters, outdoor transformer, transmission towers, shield wire supports and lightning protection system down conductors. However, it is not necessary to install ground rods closer together than 6 meters. When required by the presence of rock, a hole shall be bored and backfilled with bentonite before installing the ground rod.

Ground rod shall be installed on a ring of ground-conductor surrounding the building and connected to the building grounding- system. The ring shall be located about one (1) meter from the building walls.

4.2.3 Connections

All buried connections shall be made by exothermic welding. Heavy duty molds and powder cartridges shall be used. The Manufacturer's instruction and safety precautions shall be strictly adhered to.

Connections made above grade may be by exothermic welding or heavy duty compression sleeves. Connections to equipment shall be made by means of lugs or by exothermic welding.

4.3 Objects to be connected to the Grounding System

All above ground metallic objects such as transformers, tanks, transmission towers, metal framing, long runs of exposed piping and outdoor mechanical equipment shall be connected to the grounding system in at least two (2) places. Long objects shall be connected to the grounding system at 30 meters intervals. At least one (1) of the connections to outdoor mechanical equipment shall be to the motor ground pad. Ground conductors shall be run in the immediate vicinity of the object to limit the touch voltages to safe values.

All fences shall be included in the grounding system, except where fence isolations are called for in the grounding drawings. A ground conductor shall be run 1.0 to 1.3 meters outside of the fence. The fence posts shall be connected to the grounding system at intervals of about thirty (30) meters for property fences. A ground conductor shall be run outside the swing of all gates.

4.3.1 Disconnecting Switches

A protective grid shall be installed under all disconnecting switch and grounding switches operators, so that a person must stand above the grid at all times when operating the switch. The protective grid shall be connected both to the grounding system and to the disconnect switch operator.

4.3.2 Ground Cover

The substation ground cover shall be in accordance with the applicable civil/structural specifications substation ground cover.

4.3.3 Shield Wires

All shield wires in the substation extension area above the 500 kV connections structures shall be connected to the existing 500 kV structures. The conductivity of the shield wires that connected from 500 kV substation to the power plant

take-off gantry shall not be less than that of a 240 mm² copper cable.

5. PERFORMANCE TESTING AND GUARANTEES

All exothermic welds shall be inspected before the joints are sealed.

The continuity of ground cables shall be verified if there is a possibility that they could have been damaged by excavation, etc.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for outdoor and substructure grounding shall conform to the requirements included in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control documentation as specified shall include but not be limited the following:

- (1) Design Calculations.
- (2) Procedures for making and inspecting exothermic welds and the installation of compression fittings.
- (3) Procedures for sealing joints in lead covered cable and in underground connections.
- (4) Procedures for ensuring the integrity of the grounding system where excavation has occurred in an area after the grounding system has been installed.
- (5) Installation Inspection Reports.
- (6) Installation Test Reports.

(Last page of E-240)